



Analyzing the Correlation Between Ski Pole Design and Buckling

Mari Zampa

Introduction

- This project sought to investigate how ski pole design and material usage impacts maximum buckling load.
- Given our group member’s experience with cross country skiing, we hope to better understand how forces are propagated throughout different ski pole designs.
- We designed, calculated, prototyped, ran an FEA, and tested two different ski pole designs to analyze how critical buckling forces differ in different ski pole designs and materials.



Figure 1: Ski pole design variation

Life Sized Model

Prototyping:

- Used CAD to create life sized models of two different ski pole variations.

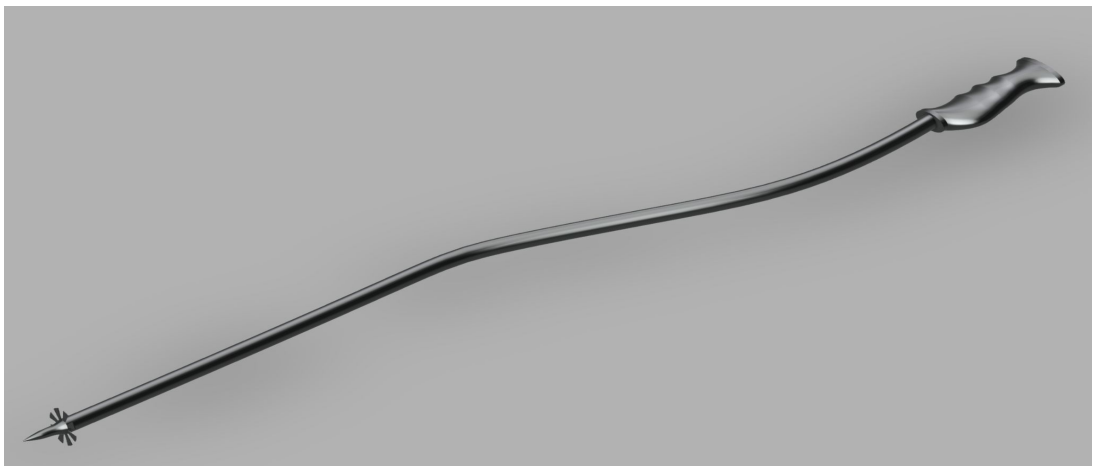


Figure 2: Bent Pole CAD

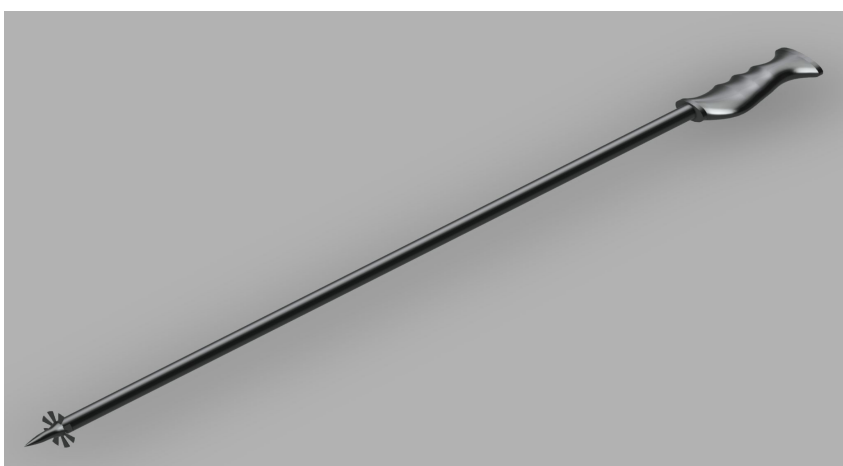


Figure 3: Straight Pole CAD

Hand Calculations:

- We then performed a basic hand calculation to estimate the critical force to produce buckling in the straight ski pole.

Given:

- Length: 1.2 m
- Diameter: 2 cm
- Wall Thickness: 2.5 mm
- Material: **Aluminium 6061**
- K = 2

Governing Equation:
$$F = \frac{\pi^2 EI}{(KL)^2}$$

Results:

- **Straight pole: 634.89 N**

Finite Element Analysis:

- We then utilized Fusion 360 to better model the critical loads and buckling we would expect within our ski pole.
- Used a fixed bottom and free top, applying 1 N of force downward to the top
- Fusion multiplier was used to determine critical force

Results:

- **Straight pole: 636.345 N**
- **Bent pole: 564.993 N**

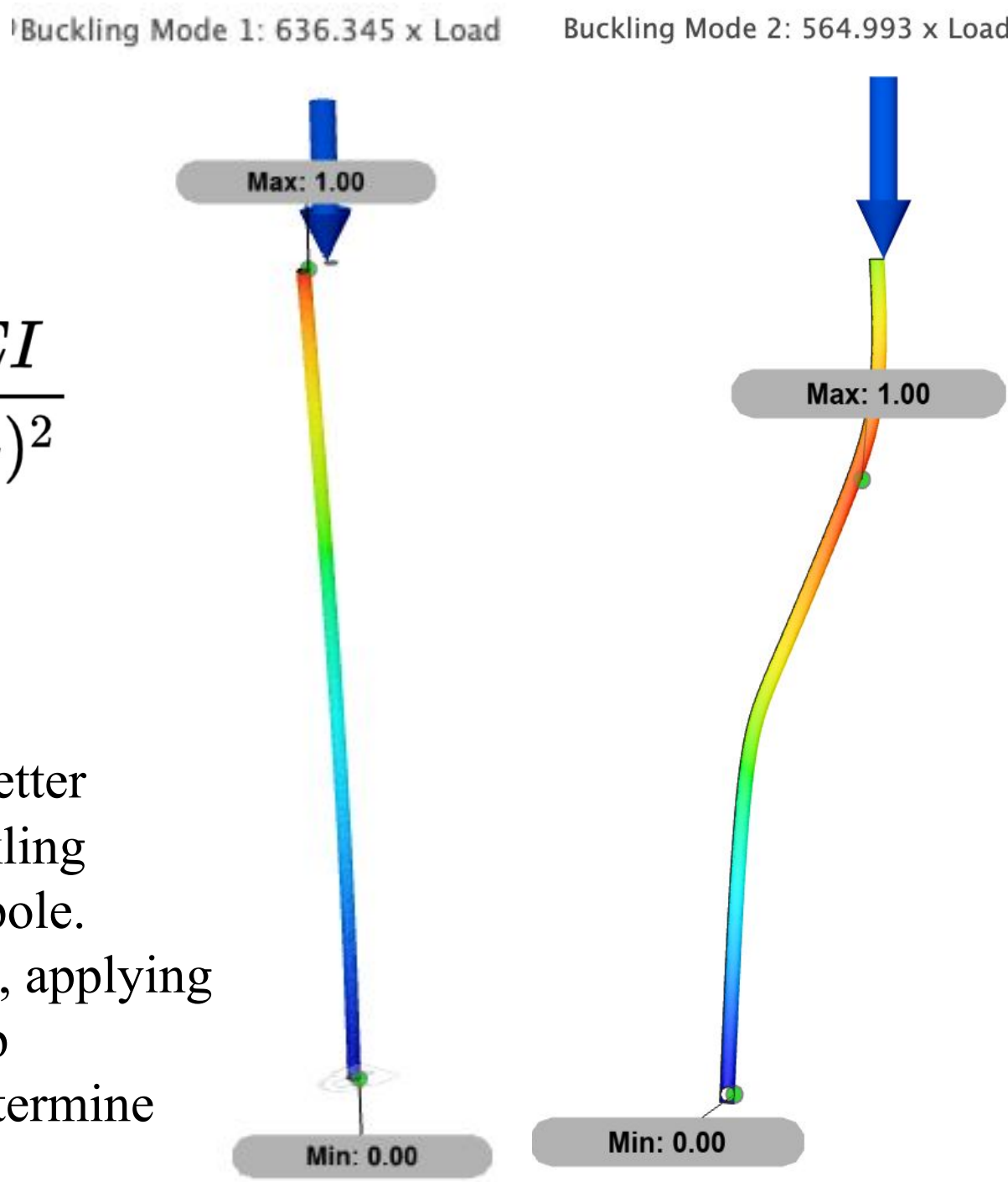


Figure 4: (Left) Scale, Straight Pole FEA; (Right) Scale, Bent Pole FEA

Test Model

Prototyping:

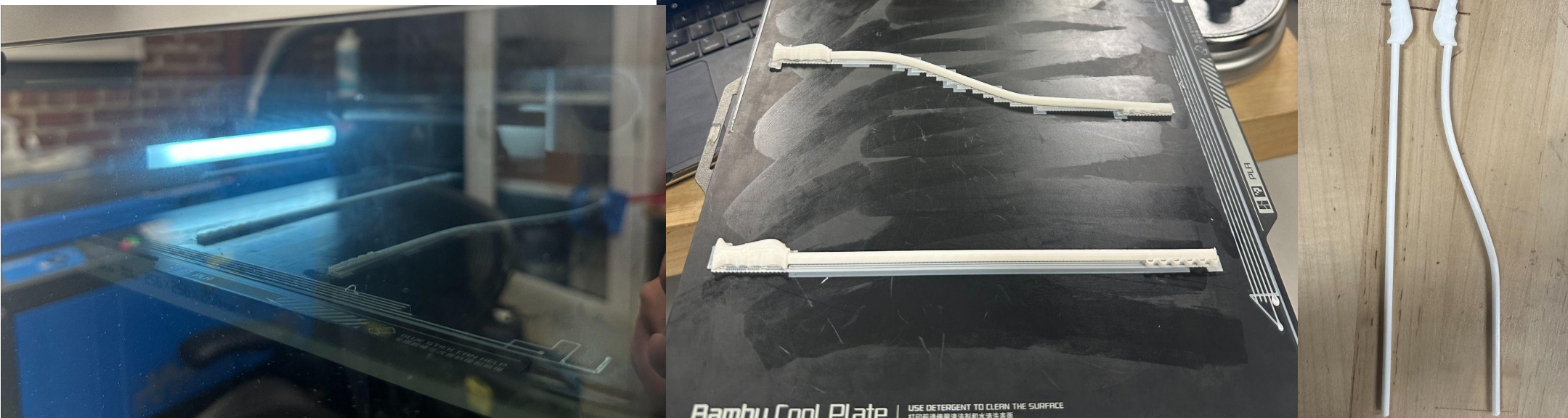


Figure 5: 3D Printing Process

- To scale down our model for testing, we used a scale factor of 0.1833 which maximized the size of our poles given the 3D printer
- These prototypes were printed without a pointed tip and flange for simplicity during testing

Hand Calculations:

- We then redid our hand calculations to estimate the amount of force expected to induce buckling on the straight rod.

Given:

- Length: 22 cm
- Diameter: 3.8 mm
- Wall Thickness: 0.475 mm
- Material: **Polylactic acid (PLA)**
- K = 1.5

Finite Element Analysis:

- We again used Fusion 360 to better model the critical loads and buckling expected in our tests
- Used a pinned bottom and free top to model test conditions, applying 1 N of force downward to the top
- Fusion multiplier was used to determine critical force

Physical Testing:

- Our setup included a scale, a piece to hold the base of the pole, a clamp, and the scaled prototypes.
- The clamp was then tightened until the prototypes visually buckled and the mass was noted.

Table 1: Analysis Results

Pole Design	Hand Calculations	FEA	Testing
Straight Pole	2.09 N	1.952 N	1.87 N
Bent Pole	n/a	1.165 N	1.28 N

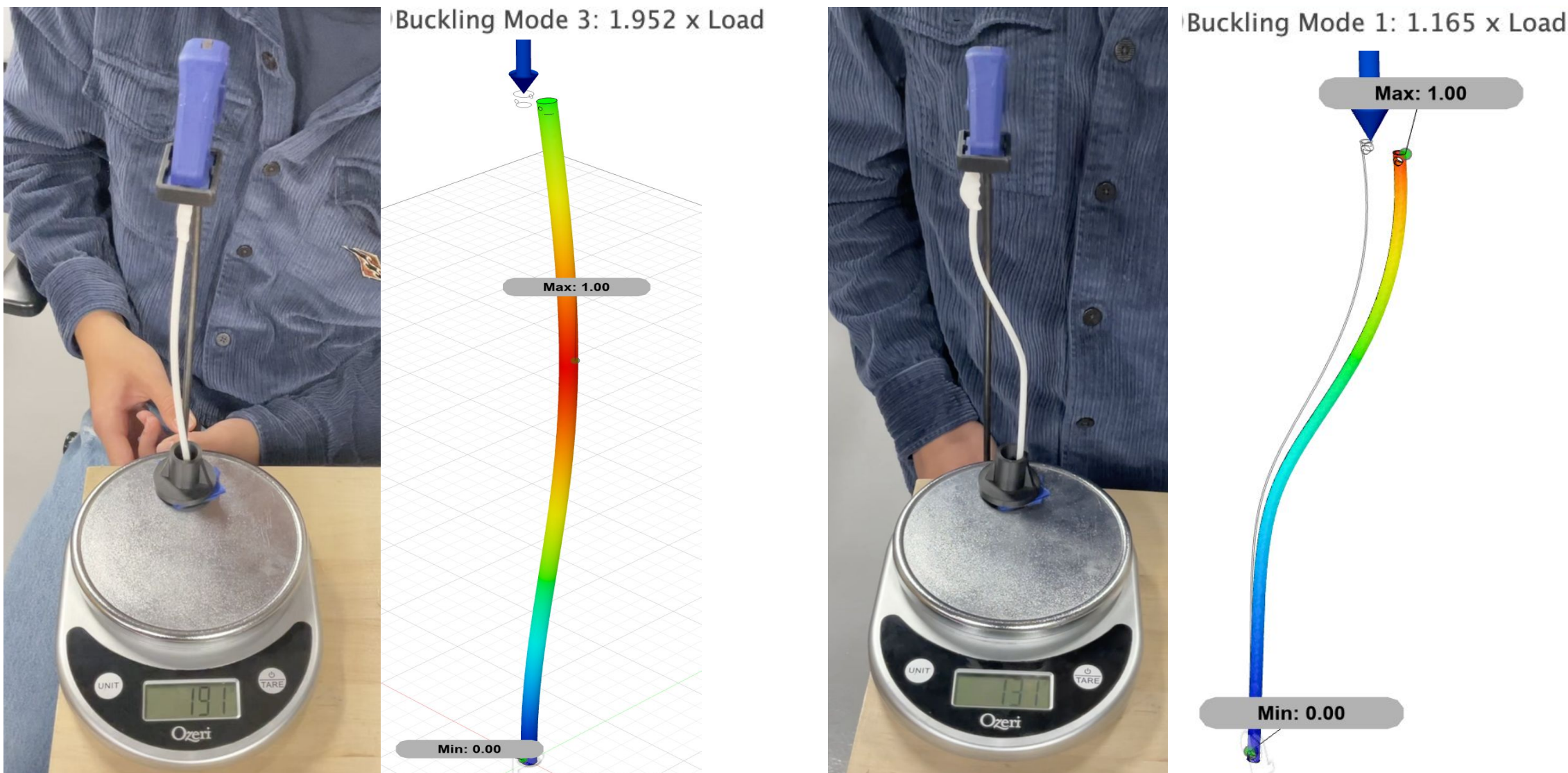


Figure 6: (Left-Right) Straight pole test; Straight pole scaled FEA; Bent pole test; Bent pole scaled FEA

Assumptions and Errors

- Generous assumptions were made for both the hand calculations and FEA in order to more easily obtain values.
 - FEA ran with polystyrene [similar material traits to PLA; E = 3.2 GPa (polystyrene) vs E = 3.3 GPa (PLA)]
 - Assumed material was uniform despite being 3D Printed
 - Estimated 1.5 K-value (between pinned-pinned and fixed-free setup)
 - Hand-calculations were not done for the bent pole due to the shape’s complexity
- Testing was very rudimentary:
 - Undefined boundary conditions (pinned vs fixed vs free)
 - Pressure was increased in spurts
 - Scale did not continuously calibrate (leading to jumps in readings)
 - Buckling was determined visually

Discussion and Conclusion

- Our results showed that the **straight ski pole consistently could withstand more force before buckling compared to the bent ski pole.**
- When looking at the results, the straight pole’s critical buckling force averaged at **1.95 N**, while the bent pole’s critical buckling force averaged at **1.22 N**. For the straight pole, our testing force was less than the hand calculation and FEA force. However, for the bent pole, our testing force was greater than our FEA force. By comparing all three methods, it allowed us to gather more data to confirm our findings.
- These differences may be attributed to our idealized K-value in our hand calculations and our use of Polystyrene instead of PLA in our FEA.
- Additionally, hand calculations and FEA assumed an idealized ski pole (perfect geometry, ideal boundary conditions, purely axial load). Physical testing, however, involved an actual pole which has imperfections and real-life variabilities. All these aspects combined give an actual buckling load that is different than the ideal buckling load in both designs (both higher and lower).
- Upon further research, these findings make sense as the curved design of the ski poles is actually intended for more practicality in slalom skiing where the poles need to wrap around the skier’s body for more efficiency.



Figure 7: Slalom Skier

Future Testing

- Future testing would require that we develop a more sophisticated fabrication process in order to more accurately test our real life models.
- With this functionality, we could add to this experiment, pairing different materials with designs to more accurately access the optimal goal.